

Designing a Digital Body Reshaping Mirror for Body Image Intervention

ALISSA TSAY SHRUTI MAHAJAN (ADVISOR)
COMPUTER SCIENCE DEPARTMENT, UNION COLLEGE



ABSTRACT

TEX This project presents a Digital Body Reshaping Mirror designed to support body-image therapy and more. Our aim was to design a mirror that can visualize a user in a requested weight in real-time. The goal was to make the visualization realistic, while keeping the system light so it could run on any software. Upcoming user studies will assess believability, emotional response, and usability to explore whether its usability and further areas for improvement.

INTRODUCTION, MOTIVATION, AND BACKGROUND

Anorexia nervosa is an eating disorder characterized by an intense fear of gaining weight and a distorted body image. One clinically approved method to help patients confront these fears is mirror exposure therapy, which involves repeated exposure to one's own reflection [3, 4]. This project builds on that idea by exposing patients to a reflection with a slightly higher body weight, aiming to strengthen the therapeutic effect. The goal is to develop a lightweight, low-cost system capable of visualizing a person at a higher body weight in real time. The software takes a live camera feed and modifies each frame to increase key body areas, projecting the adjusted image back to the user without noticeable delay.

The visualization of weight gain is grounded in research on physiological fat distribution and clinical patterns of weight restoration in pre-menopausal women and women with anorexia nervosa, who primarily regain fat in the thighs and, to a lesser extent, in the abdomen [5, 7]. These findings form the anatomical foundation of my approach, ensuring that visualized weight changes appear plausible and natural.

Several research efforts have explored the visualization and modification of human body shape through digital or augmented interfaces. Richter et al. introduced a full-body reshaping system using RGB-D cameras for real-time modification of parameters such as body weight and muscle tone [9]. Fiedler et al. developed augmented-reality mirrors designed for daily self-reflection on body image [6], while Cornelissen et al. used 3D body scans to simulate realistic weight gain in anorexia-spectrum patients, revealing that overestimation of body size is most pronounced in the hips and thighs [2]. Building on these works, this project strives to create a similarly meaningful and believable experience through a simpler, more accessible real-time system.



FIGURE 1:Types of fat distribution [8]

QUESTION

How can we design a Digital Body Reshaping Mirror to aid patients with eating disorders?

SYSTEM DESIGN

I made my mirror using *MediaPipe Pose Tracking*, *OpenCV*, and *NumPy* python libraries. The system warps the pixels around the hips and abdomen to simulate weight gain.

Process Overview

Pose Detection: MediaPipe Pose joint locations; the average of hip joint positions defines the warp center $(u_{CenterX}, u_{PeakY}) \in [0, 1]$.

Warp Mapping: A Gaussian-based scale function defines vertical dependence of horizontal expansion:

$$scale(y) = 1 + u_{Gain} \cdot e^{-\left(\frac{(y-u_{PeakY})^2}{\sigma_y^2}\right)}$$

Pixel coordinates are remapped as:

$$x' = u_{CenterX} + \frac{(x - u_{CenterX})}{scale(y)}, \quad y' = y$$

producing remap grids (map_x, map_y) for `cv2.remap()`.

Segmentation & Compositing: The warped and mirrored frame is segmented and layed over a captured background:

$$I_{out} = M \cdot I_{person} + (1 - M) \cdot I_{bg}$$

where M is a blurred binary mask.

Algorithm Steps

- 1 Initialize webcam, pose, and segmentation models.
- 2 Capture background.
- 3 For each frame:
 - 1 Detect pose $\rightarrow$ compute $u_{CenterX}, u_{PeakY}$
 - 2 Build warp maps using Gaussian profile
 - 3 Apply `cv2.remap()` and mirror
 - 4 Segment and composite over background
 - 5 Display until user quits



FIGURE 2:The User Studies Setup. Participants will stand in front of large screen, simulating the mirror in an enclosed room, with the experimenter setting weight parameters on laptop and observing from the outside.

USER STUDIES

The algorithm produces a realistic visualization of a person at a higher body weight, up to approximately 40% ($u_{Gain} = 0.4$) realistic weight gain. Beyond that, users begin to notice inconsistencies between the warped abdomen and hips and the unchanged regions such as the hands, calves, and face.

I plan to conduct participant studies to determine the threshold at which the deformation begins to appear unrealistic. Each participant will first be exposed to their reflection at a 30% simulated weight gain and asked to rate the believability of the experience, the persistence of glitches or other distracting elements, the emotional impact of the experience, and their willingness to use such a mirror in the future. Figure 2 illustrates the setup of the studies. Additionally, participants will view their reflection at 10%, 20%, 30%, 35%, and 40% simulated weight gain and estimate the perceived amount of weight gain after each exposure.

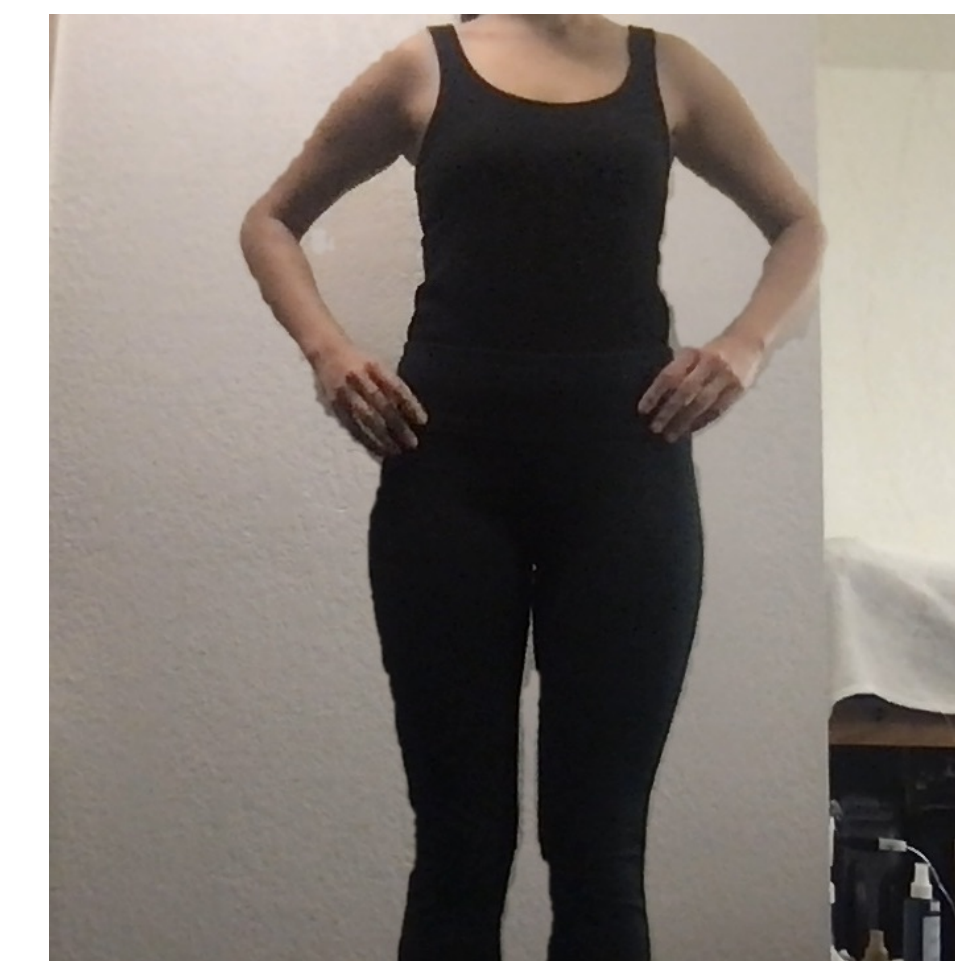


FIGURE 3:*

(a) Original



FIGURE 4:*

(b) Modified, 30% weight gain

FIGURE 5:Front images with modification and without side by side

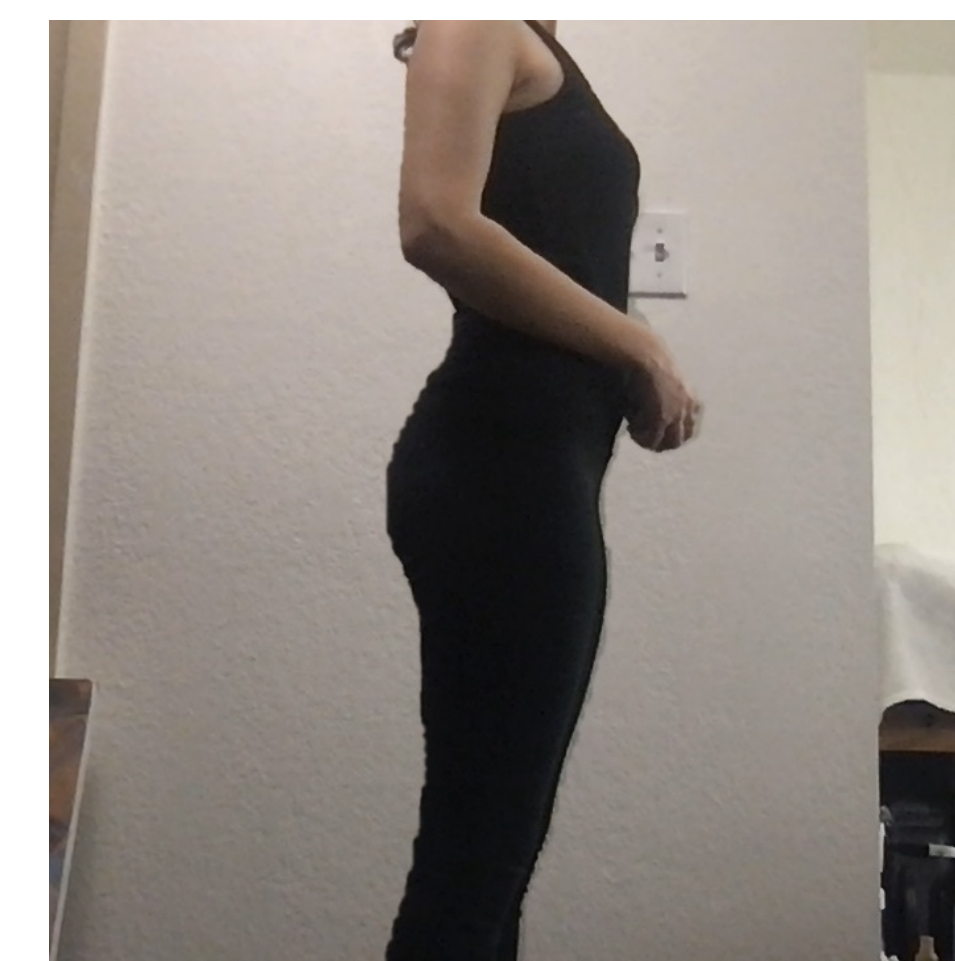


FIGURE 6:*

(c) Original

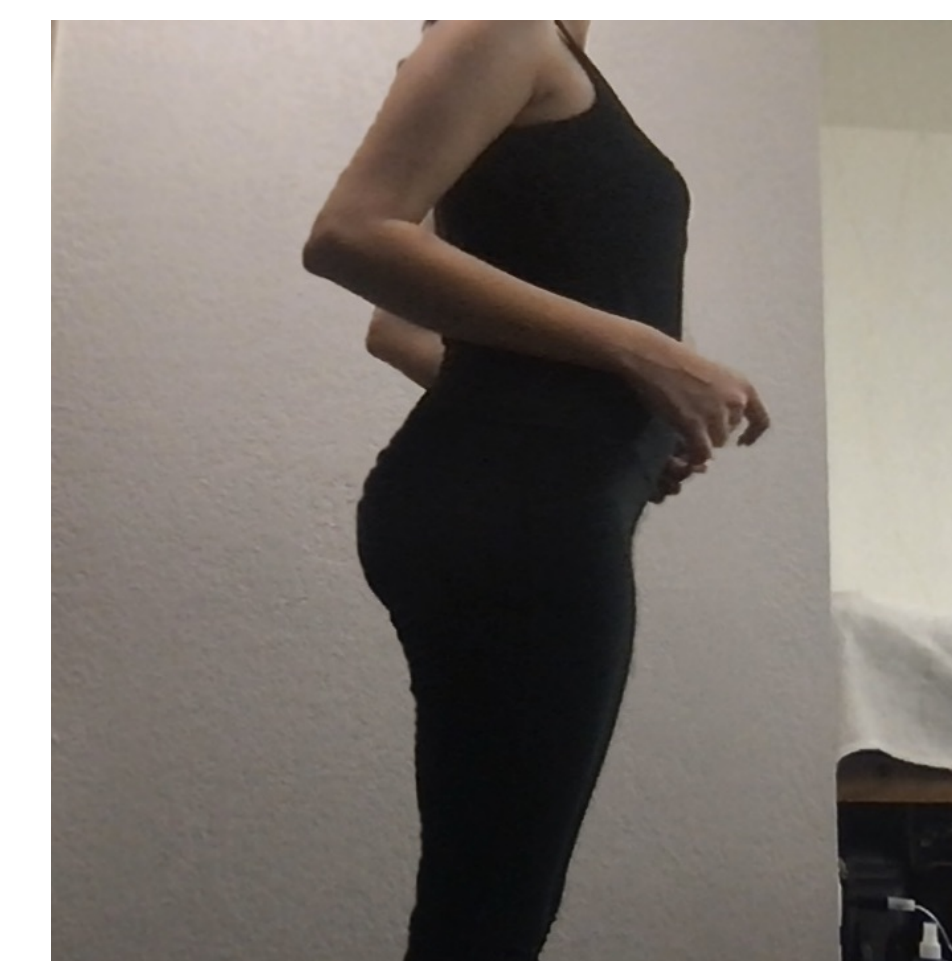


FIGURE 7:*

(d) Modified, 30% weight gain

FIGURE 8:Side images with modification and without side by side

DATA ANALYSIS & RESULTS

SUS scores ($M = 63.0$) indicated **marginally acceptable to good** usability [1]. Participants strongly recognized the reflected body as their own ($M = 4.00$, $SD = 0.71$), and perceived weight gain rose sharply only at higher distortion levels, with realism remaining consistently high throughout ($r = 0.18$).

CONCLUSIONS

Overall, the goal of developing a light weight system that can visualize a given person at a higher body weight, with an opportunity to set the weight gain and body shape parameters in real time was achieved. My software runs on any computer or laptop, however, older laptops might experience some glitching or delays. While my research could raise ethical or legal concerns, the approval of several councils will be obtained prior to conducting clinical studies on ED patients. My participants will also be provided with full information on the nature of the experiment, its potential dangers as well as made aware of their ability to withdraw at any moment. Given more time, I would like to work on integrating the visualization of calves, hands, and most of all face at a higher weight, which will make the visuals more believable and the system more versatile.

ACKNOWLEDGMENTS

I would like to thank my adviser, Shruti Mahajan, for supporting me throughout this research. I would like to thank professor Walker of the psychology department who came up with the body weight modification idea, as well as Mai Tran, and Lauren Levitt for prior psychology research. I would like to thank all aforementioned for continuous feedback and willingness to participate in my studies. I would also like to thank Undergraduate Research for granting me an SRG fund for necessary equipment and participant payments for this study.

REFERENCES

- [1] BROOKE, J. Sus: A “quick and dirty” usability scale, 1996.
- [2] CORNELISSEN, K. K. E. A. Body size estimation in women with anorexia nervosa and controls. *Sci. Rep.* (2017).
- [3] DELINSKY, S. S., AND WILSON, G. T. Mirror exposure for the treatment of body image disturbance. *Int. J. Eat. Disord.* (2006).
- [4] DÍAZ-FERRER, S. E. A. Pure vs guided mirror exposure in women with high body dissatisfaction. *Eating Disorders* (2015).
- [5] FENTON, A. E. A. Weight, shape, and body composition changes at menopause. *J. Mid-Life Health* (2021).
- [6] FIEDLER, M. L. E. A. Holographic ar mirrors for daily self-reflection. In *CHI '23 Workshop* (2023).
- [7] GRINSPOON, S. E. A. Regional fat distribution and effects of estrogen in women with anorexia nervosa. *Am. J. Clin. Nutr.* (2005).
- [8] MARCADENTI, A., AND OLIVEIRA DE ABREU-SILVA, E. Different adipose tissue depots: Metabolic implications and effects of surgical removal. *Endocrinología y Nutrición (English Edition)* (2015).
- [9] RICHTER, M. E. A. Real-time reshaping of humans. In *3DIMPVT Conf.* (2012).